

A trace-vanishing dichotomy for the Iwahori–Hecke algebra of the symmetric group

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Abstract

For partitions $\lambda = (\hat{\lambda}, 1^q)$ with $\ell(\hat{\lambda}) \geq 2$ and every row of $\hat{\lambda}$ of length ≥ 2 , we exhibit a specific element $\Omega^{(\lambda)}$ of the Iwahori–Hecke algebra $H_q(S_n)$ whose trace on the Specht module V^λ obeys a clean combinatorial dichotomy: $\mathrm{tr}_{V^\lambda}(\Omega^{(\lambda)}) \equiv 0$ as a rational function of q if and only if a combinatorial threshold $\tau(\hat{\lambda}) := \max(0, q + 2\ell(\hat{\lambda}) - 1 - |\hat{\lambda}|)$ is at least 1. The proof combines four ingredients: (i) a *multi-row sign-kill meta-theorem* (Pillar 1) placing the image of $\Omega^{(\lambda)}$ in $\bigcap_j E_j^-$; (ii) a *right-absorption identity* $\Omega^{(\lambda)}(T_1 + 1) = (q + 1)\Omega^{(\lambda)}$; (iii) *per-SYT positivity* of every diagonal entry of $\Omega^{(\lambda)}$ in the asymmetric Hoefsmit basis; and (iv) an explicit *two-band SYT construction* T_{uniform} exhibiting a non-vanishing diagonal entry when $\tau(\hat{\lambda}) = 0$. The dichotomy is verified for all 5957 shapes with $|\lambda| \leq 25$. We close with a conjectural categorical lift in the homotopy category of Soergel bimodules.

Contents

1	Introduction	2
2	Preliminaries	3
2.1	The Iwahori–Hecke algebra	3
2.2	The Hoefsmit asymmetric seminormal basis	3
2.3	Descent eigenspaces	4
3	The element $\Omega^{(\lambda)}$ and the threshold τ	4
3.1	Definition of $\Omega^{(\nu)}$	4
3.2	The threshold τ	4
3.3	Statement of the main theorem	5
4	Forward direction: structural nilpotence	5
4.1	Pillar 1: the multi-row meta-theorem	5
4.2	Right-absorption	6
4.3	The trace-vanishing corollary	6
5	Converse: per-SYT positivity	6
5.1	Hoefsmit positivity	7
5.2	Reduction to one non-vanishing diagonal	7
5.3	Descent-1 kill and the stay-walk lower bound	7

6	Converse: existence of T^*	8
6.1	The construction T_{uniform}	8
6.2	Three elementary lemmas	9
6.3	The main existence theorem	9
7	Sharpness at $\tau + 1$	9
7.1	Statement	9
7.2	The one-fact proof	10
7.3	Hook base case	10
7.4	Verification	11
8	Per-SYT diagnostics	11
8.1	Diagnostic 1: per-SYT positivity	11
8.2	Diagnostic 2: leftmost-factor SC vanishing	11
8.3	Diagnostic 3: Pillar-1-side SR vanishing	11
8.4	Open: companion diagnostic and interior zeros	12
9	Computational verification	12
10	Outlook: a categorical lift	13
10.1	The conjecture	13
10.2	What the lift should achieve	13
10.3	Reading list	13
10.4	The Demazure–Lusztig bridge	14
10.5	What this paper does not address	14

1 Introduction

When does the trace of a non-central element of the Iwahori–Hecke algebra vanish identically as a function of q ? For most algebra elements this question is meaningless or trivial. For a specific family $\{\Omega^{(\lambda)}\}$ indexed by partitions λ with at least two rows and a column-1 tail, we exhibit a clean combinatorial dichotomy.

The element $\Omega^{(\lambda)}$ is built out of products of $S_j := T_j + 1$. For $\nu \vdash N$ with $\ell = \ell(\nu)$, set $R'_k := S_{k-1}S_{k-2} \cdots S_1$ and

$$\Omega^{(\nu)} := R'_{\ell+1} R'_{\ell+2} \cdots R'_N.$$

For $\widehat{\lambda}$ with $\ell(\widehat{\lambda}) \geq 2$ rows, all of length ≥ 2 , and $\lambda = (\widehat{\lambda}, 1^q)$, the threshold

$$\tau(\widehat{\lambda}) := \max(0, q + 2\ell(\widehat{\lambda}) - 1 - |\widehat{\lambda}|)$$

encodes how much the tail length q exceeds the minimum compatible with the shape of $\widehat{\lambda}$.

Theorem 1.1 (Main dichotomy). *Let $\widehat{\lambda}$ be a partition with $\ell(\widehat{\lambda}) \geq 2$ rows, every row of length ≥ 2 , and let $\lambda = (\widehat{\lambda}, 1^q) \vdash n$ for some $q \geq 1$. Then, as an identity of rational functions of q ,*

$$\text{tr}_{V^\lambda}(\Omega^{(\lambda)}) \equiv 0 \iff \tau(\widehat{\lambda}) \geq 1.$$

Equivalently, the trace vanishes identically iff $q \geq |\widehat{\lambda}| - 2\ell(\widehat{\lambda}) + 2$. The dichotomy has been verified on all 5957 shapes $|\lambda| \leq 25$ satisfying the row-length hypothesis.

Structure of the paper. Section 2 fixes notation: the Iwahori–Hecke algebra $H_q(S_n)$, the asymmetric Hoefsmit basis with $b' = 1$, and the descent eigenspaces E_j^+, E_j^- . Section 3 defines $\Omega^{(\lambda)}$ and τ , and tabulates first examples. Section 4 proves the forward direction ($\tau \geq 1 \Rightarrow \text{tr} = 0$) from Pillar 1 and right-absorption. Sections 5 and 6 prove the converse via per-SYT positivity and an explicit construction T_{uniform} . Section 7 proves strong sharpness $B^{(\lambda)} \cap E_{\tau+1}^- = \{0\}$, pinning the descent depth at exactly τ . Section 8 collects three independent per-SYT diagnostics for τ . Section 9 records the scope of computational verification. Section 10 states a conjectural categorical lift and the Demazure–Lusztig bridge to K -theory of the flag variety.

Related work. The seminormal basis of $H_q(S_n)$ goes back to Hoefsmit [1] and Murphy [2]; the cellular structure is due to Dipper–James [3]. Ram [4, 5] developed Markov traces and Jucys–Murphy product machinery in this setting. Tolmachov–Zhylynskiy [6] compute traces of products of polynomials in Jucys–Murphy elements in closed form; their formula motivates but does not directly settle the dichotomy because $\Omega^{(\lambda)}$ is not a central JM product (see Remark 9.1). The categorical home of our dichotomy is the derived horizontal trace $\text{HH}_*(\text{SBim}_n)$ studied by Gorsky–Hogancamp–Wedrich [7]; the relevant Specht-isotypic projectors P_λ are Hogancamp’s categorified Young symmetrizers [8, 9]. Recent work of Trinh–Williams [10] on Hecke trace formulas via partial Springer resolutions is, to our knowledge, the only published reference citing Tolmachov–Zhylynskiy.

Two routes to per-SYT positivity were investigated and *deflated*. The first, Goertzen–Williamson’s variational characterization of the seminormal basis [13], gives the right structural picture at $v = 1$ but does not transfer to generic q . The second, Bai–Gu–Guo–Liu’s 1423-avoidance criterion for non-negativity in certain pipe-dream rings [14], matches our diagonals on small shapes but is a different ring; the apparent bridge does not survive scrutiny. We comment on both in Section 10.

2 Preliminaries

2.1 The Iwahori–Hecke algebra

Let $H_q(S_n)$ denote the Iwahori–Hecke algebra of the symmetric group S_n over the field $\mathbb{Q}(q)$, with generators $T_1, \dots, T_{n-1}$ and relations

$$T_i^2 = (q - 1)T_i + q, \quad T_i T_{i+1} T_i = T_{i+1} T_i T_{i+1}, \quad T_i T_j = T_j T_i \text{ for } |i - j| \geq 2.$$

The eigenvalues of each T_j on any finite-dimensional module are q and -1 . Set

$$S_j := T_j + 1, \quad j = 1, \dots, n - 1.$$

Lemma 2.1 (Square identity). $S_j^2 = (q + 1)S_j$ in $H_q(S_n)$ for every j .

Proof. $(T_j + 1)^2 = T_j^2 + 2T_j + 1 = ((q - 1)T_j + q) + 2T_j + 1 = (q + 1)T_j + (q + 1) = (q + 1)(T_j + 1)$. $\square$

Equivalently, $\frac{1}{q+1}S_j$ is an idempotent (the projector onto the $+q$ -eigenspace of T_j).

2.2 The Hoefsmit asymmetric seminormal basis

For $\nu \vdash N$, the Specht module V^ν has the Hoefsmit seminormal basis $\{v_T : T \in \text{SYT}(\nu)\}$ in which each T_j acts by an explicit two-dimensional block on each pair $\{T, s_j T\}$. We use the *asymmetric* normalization with $b' = 1$. Recall the content $c(r, k) := k - r$ of cell (r, k) and the axial distance $d_j(T) := c_{j+1}(T) - c_j(T)$ from j to $j + 1$. The action of T_j on v_T is

- (a) $d_j(T) = 1$ (same row): $T_j v_T = q v_T$.
- (b) $d_j(T) = -1$ (same column): $T_j v_T = -v_T$.
- (c) $|d_j(T)| \geq 2$ (so $s_j T \in \text{SYT}(\nu)$):

$$\begin{aligned} T_j v_T &= a_d v_T + v_{s_j T}, \\ T_j v_{s_j T} &= (a_d a_{-d} + q) v_T + a_{-d} v_{s_j T}, \end{aligned}$$

where $d = d_j(T)$ and $a_d := (q-1)q^d/(q^d-1)$.

2.3 Descent eigenspaces

For each $j \in \{1, \dots, N-1\}$, the operator T_j has eigenvalues $\{q, -1\}$ on V^ν , giving the orthogonal decomposition

$$V^\nu = E_j^+(V^\nu) \oplus E_j^-(V^\nu),$$

where

$$E_j^+ := \ker(T_j - q)|_{V^\nu}, \quad E_j^- := \ker(T_j + 1)|_{V^\nu}.$$

Equivalently, S_j acts on E_j^+ by the scalar $q+1$ and on E_j^- by 0. In the Hoefsmit basis, T_1 is diagonal: $T_1 v_T = q v_T$ if T has entries 1, 2 in the same row (*descent-0 at 1*), and $T_1 v_T = -v_T$ if in the same column (*descent-1 at 1*).

Remark 2.2 (Notational overload). The symbol q plays two distinct roles in this paper. Throughout, q is both the formal Hecke parameter (so that H_q is defined over $\mathbb{Q}(q)$) and the tail length of $\lambda = (\widehat{\lambda}, 1^q)$. This coincidence reflects the framework in which the four source papers were developed and we keep it; context disambiguates. The principal results, like Theorem 1.1, are about the trace as an identity of rational functions in q at the specific integer value q chosen as the tail length.

3 The element $\Omega^{(\lambda)}$ and the threshold τ

3.1 Definition of $\Omega^{(\nu)}$

For $\nu \vdash N$ with $\ell = \ell(\nu)$, set

$$R'_k := S_{k-1} S_{k-2} \cdots S_1 = (T_{k-1} + 1)(T_{k-2} + 1) \cdots (T_1 + 1), \quad R'_1 := 1,$$

and define

$$\Omega^{(\nu)} := R'_{\ell+1}^{(\nu)} R'_{\ell+2}^{(\nu)} \cdots R'_N^{(\nu)} \in H_q(S_n).$$

We will write $\Omega = \Omega^{(\lambda)}$ when λ is fixed.

3.2 The threshold τ

Throughout the remainder of the paper, $\widehat{\lambda}$ denotes a partition with $\ell(\widehat{\lambda}) \geq 2$ rows, every row of length ≥ 2 , and we set

$$\lambda := (\widehat{\lambda}_1, \dots, \widehat{\lambda}_{\ell(\widehat{\lambda})}, \underbrace{1, \dots, 1}_q) \vdash n, \quad n = q + |\widehat{\lambda}|,$$

with $\ell(\lambda) = \ell(\widehat{\lambda}) + q$. The threshold is

$$\tau(\widehat{\lambda}) := \max(0, q + 2\ell(\widehat{\lambda}) - 1 - |\widehat{\lambda}|).$$

Equivalently, writing $|\widehat{\lambda}| - 2\ell(\widehat{\lambda}) = \sum_r (\widehat{\lambda}_r - 2)$ for the number of cells of $\widehat{\lambda}$ outside its first two columns,

$$\tau(\widehat{\lambda}) = \max(0, q - 1 - (|\widehat{\lambda}| - 2\ell(\widehat{\lambda}))).$$

The condition $\tau(\widehat{\lambda}) \geq 1$ is equivalent to $q \geq |\widehat{\lambda}| - 2\ell(\widehat{\lambda}) + 2$.

Example 3.1. We tabulate small cases.

$\widehat{\lambda}$	$\tau(\widehat{\lambda})$	smallest q with $\tau \geq 1$
(2, 2)	$\max(0, q - 1)$	$q = 2$
(3, 2)	$\max(0, q - 2)$	$q = 3$
(3, 3)	$\max(0, q - 3)$	$q = 4$
(2, 2, 2)	$\max(0, q - 1)$	$q = 2$
(3, 2, 2)	$\max(0, q - 2)$	$q = 3$

The smallest non-trivial dichotomy instance is $\widehat{\lambda} = (2, 2)$ between $q = 1$ ($\tau = 0$, $\text{tr} \neq 0$) and $q = 2$ ($\tau = 1$, $\text{tr} = 0$).

3.3 Statement of the main theorem

We restate Theorem 1.1 for reference.

Theorem 3.2 (Main dichotomy, restated). *With hypotheses on $\widehat{\lambda}$ as above and $\lambda = (\widehat{\lambda}, 1^q)$, $q \geq 1$,*

$$\text{tr}_{V^\lambda}(\Omega^{(\lambda)}) \equiv 0 \text{ in } \mathbb{Q}(q) \iff \tau(\widehat{\lambda}) \geq 1.$$

The forward direction is proved in Section 4; the converse in Sections 5–6.

4 Forward direction: structural nilpotence

The forward direction follows from two ingredients: a structural containment of the image ΩV^λ inside E_1^- (Pillar 1), and an algebraic identity in the ambient Hecke algebra (right-absorption). Together they show that, in the descent-at-1 block decomposition of V^λ , the matrix of $\Omega^{(\lambda)}$ is strictly lower triangular — in particular nilpotent of order at most 2, with vanishing trace.

4.1 Pillar 1: the multi-row meta-theorem

Theorem 4.1 (Pillar 1, multi-row meta-theorem). *Let $\widehat{\lambda}$ have $\ell(\widehat{\lambda}) \geq 2$ rows, every row of length ≥ 2 , and let $\lambda = (\widehat{\lambda}, 1^q)$ with $q \geq |\widehat{\lambda}| - 2\ell(\widehat{\lambda}) + 2$ (equivalently $\tau(\widehat{\lambda}) \geq 1$). Then*

$$\Omega^{(\lambda)} V^\lambda \subseteq \bigcap_{j=1}^{\tau(\widehat{\lambda})} E_j^-|_{V^\lambda}.$$

Proof sketch. Strong induction on $|\widehat{\lambda}|$. The two key ingredients are: (i) a *threshold-matching identity* (multi-row form) showing that every length-preserving contributor $\mu_{k,*}$ obtained from the E_{n-1}^+ decomposition preserves $\tau_0(\widehat{\lambda}) := q + 2\ell(\widehat{\lambda}) - 1 - |\widehat{\lambda}|$ exactly, in both the *regular* ($\widehat{\lambda}_k \geq 3$) and *shrinking* ($\widehat{\lambda}_k = 2$) cases; (ii) a *support rule*: if every T in the support of a subspace has $p(T) \geq P$ (where $p(T)$ is the column-1 identity-prefix length), then the subspace lies in $\bigcap_{j \leq P-1} E_j^-$. Vanisher pieces (corner-pair and same-row) inflate τ_0 by 2 and are killed by the IH plus an outer R'_ℓ factor whose rightmost S_1 annihilates the IH-provided E_1^- piece. Base case $\widehat{\lambda} = (2, 2)$ at $q \geq 2$ matches the May-14 two-row meta-theorem. See [15, Theorem 1] for the complete proof. $\square$

4.2 Right-absorption

Lemma 4.2 (Right-absorption). *As an element of $H_q(S_n)$,*

$$\Omega^{(\lambda)}(T_1 + 1) = (q + 1) \Omega^{(\lambda)}.$$

Proof. By Lemma 2.1, $(T_1 + 1)^2 = (q + 1)(T_1 + 1)$. The rightmost factor of $\Omega^{(\lambda)}$ is $R'_n = (T_{n-1} + 1) \cdots (T_1 + 1)$, ending in $(T_1 + 1)$. Therefore

$$R'_n \cdot (T_1 + 1) = (T_{n-1} + 1) \cdots (T_2 + 1) \cdot (T_1 + 1)^2 = (q + 1) R'_n.$$

Multiplying on the left by $R'_{\ell+1} \cdots R'_{n-1}$ gives the claim. $\square$

Remark 4.3. Lemma 4.2 is an identity in $H_q(S_n)$, valid on every H_q -module. In particular, for any $v \in E_1^-$ we have $(T_1 + 1)v = 0$, hence

$$(q + 1) \Omega v = \Omega(T_1 + 1)v = 0,$$

so $\Omega v = 0$ since $q + 1$ is a unit in $\mathbb{Q}(q)$. Thus $\Omega^{(\lambda)}$ *always* kills E_1^- regardless of the value of $\tau(\hat{\lambda})$.

4.3 The trace-vanishing corollary

Corollary 4.4 (Forward direction). *Under the hypotheses of Theorem 4.1 (in particular $\tau(\hat{\lambda}) \geq 1$),*

$$\Omega^{(\lambda)^2} = 0 \quad \text{on } V^\lambda, \quad \text{and} \quad \text{tr}_{V^\lambda}(\Omega^{(\lambda)}) = 0.$$

Proof. Order the Hoefsmit basis of V^λ by descent-at-1: place descent-0 tableaux first, descent-1 tableaux second. Write the matrix $M(\Omega)$ of $\Omega^{(\lambda)}$ in this ordering as the block matrix

$$M(\Omega) = \begin{pmatrix} A & B \\ X & D \end{pmatrix},$$

where rows/columns are split as (E_1^+, E_1^-) . We claim $A = 0$, $B = 0$, $D = 0$.

- For T descent-0 (i.e. $v_T \in E_1^+$): by Theorem 4.1, $\Omega v_T \in E_1^-$. Hence the column of $M(\Omega)$ at T has zeros in E_1^+ rows, so $A = 0$.
- For T descent-1 (i.e. $v_T \in E_1^-$): by Remark 4.3, $\Omega v_T = 0$. So columns B and D are both zero.

We conclude

$$M(\Omega) = \begin{pmatrix} 0 & 0 \\ X & 0 \end{pmatrix},$$

and immediately $M(\Omega)^2 = 0$ and $\text{tr} M(\Omega) = 0$. $\square$

This proves the forward direction of Theorem 1.1. Note that we obtained the stronger conclusion $\Omega^2 = 0$, not merely $\text{tr}(\Omega) = 0$: the trace-vanishing dichotomy is in fact a *nilpotence* dichotomy.

5 Converse: per-SYT positivity

The converse direction requires showing that $\text{tr}_{V^\lambda}(\Omega^{(\lambda)}) \neq 0$ whenever $\tau(\hat{\lambda}) = 0$. We work in the asymmetric Hoefsmit basis with $b' = 1$ and prove that every diagonal matrix entry of $\Omega^{(\lambda)}$ lies in the non-negative cone $\mathbb{Q}_{\geq 0}(q) := \{N(q)/D(q) : N, D \in \mathbb{Z}_{\geq 0}[q]\}$. A sum of elements of this cone vanishes iff each summand does, reducing the converse to exhibiting a single non-vanishing diagonal.

5.1 Hoefsmit positivity

Lemma 5.1 (Hoefsmit positivity). *Every entry of $S_i = T_i + 1$ on V^ν in the asymmetric Hoefsmit basis ($b' = 1$) lies in $\mathbb{Q}_{\geq 0}(q)$. Explicitly, with $d = d_i(T)$:*

- (i) *Diagonal entries: $[S_i]_{T,T} = q + 1$ if $d = 1$; 0 if $d = -1$; $[d + 1]_q/[d]_q$ if $|d| \geq 2$.*
- (ii) *Off-diagonal entries (when $|d| \geq 2$): $[S_i]_{s_i T, T} = 1$ and $[S_i]_{T, s_i T} = q[d - 1]_q[d + 1]_q/[d]_q^2$. All other off-diagonals vanish.*

Proof sketch. Direct computation from the Hoefsmit action. For $|d| \geq 2$, $[S_i]_{T,T} = a_d + 1 = q^{d+1} - 1 / (q^d - 1) = [d + 1]_q/[d]_q$; for $d \leq -2$, rewrite via $[-m]_q = -q^{-m}[m]_q$ to obtain $q[k - 1]_q/[k]_q \in \mathbb{Q}_{\geq 0}(q)$. The off-diagonal computation likewise yields $q[d - 1]_q[d + 1]_q/[d]_q^2$, manifestly in $\mathbb{Q}_{\geq 0}(q)$. See [17, Lemma 1]. $\square$

Corollary 5.2 (Product positivity). $\mathbb{Q}_{\geq 0}(q)$ is closed under sums and products. Hence every entry of $\Omega^{(\lambda)} = \prod_k R'_k$ in the asymmetric Hoefsmit basis is in $\mathbb{Q}_{\geq 0}(q)$.

Theorem 5.3 (Per-SYT positivity). *For every $T \in \text{SYT}(\lambda)$, the diagonal coefficient*

$$p_T(q) := [v_T](\Omega^{(\lambda)} v_T)$$

lies in $\mathbb{Q}_{\geq 0}(q)$. Consequently,

$$\text{tr}_{V^\lambda}(\Omega^{(\lambda)}) = \sum_{T \in \text{SYT}(\lambda)} p_T(q) \in \mathbb{Q}_{\geq 0}(q).$$

Proof. Immediate from Corollary 5.2: each p_T is a diagonal entry of $\Omega^{(\lambda)}$. $\square$

5.2 Reduction to one non-vanishing diagonal

Corollary 5.4 (Trace vanishes iff every diagonal vanishes). $\text{tr}_{V^\lambda}(\Omega^{(\lambda)}) \equiv 0$ as a rational function of q iff $p_T(q) \equiv 0$ for every $T \in \text{SYT}(\lambda)$.

Proof. The cone $\mathbb{Q}_{\geq 0}(q)$ consists of rational functions taking non-negative values at every real $q_0 > 0$ that is not a pole. A finite sum of such functions vanishing identically forces each summand to vanish at every common point of positivity, hence everywhere. $\square$

To complete the converse, we need a single $T \in \text{SYT}(\lambda)$ with $p_T(q) \neq 0$ when $\tau(\hat{\lambda}) = 0$.

5.3 Descent-1 kill and the stay-walk lower bound

Lemma 5.5 (Descent-1 kill). *If T is descent-1 at position 1, then $\Omega^{(\lambda)} v_T = 0$ and hence $p_T(q) = 0$.*

Proof. The rightmost factor of $\Omega^{(\lambda)}$ is $S_1 = T_1 + 1$, and $S_1 v_T = 0$ for T descent-1 at 1 by Lemma 5.1(i) with $d = -1$. $\square$

So only descent-0-at-1 SYTs can contribute non-trivially. Define an SYT T to be *column-descent-free* if no j has j and $j + 1$ in the same column at adjacent rows (Definition 6.1 below).

Lemma 5.6 (Stay-walk lower bound). *If $T \in \text{SYT}(\lambda)$ is descent-0 at 1 and column-descent-free, then*

$$p_T(q) \geq \prod_j [S_j]_{T,T}^{e_j} > 0 \quad \text{in } \mathbb{Q}_{\geq 0}(q),$$

where e_j counts occurrences of S_j in $\Omega^{(\lambda)}$.

Proof sketch. Expand $\Omega^{(\lambda)}v_T$ as a sum over Hoefsmit walks $T = T_0 \rightarrow T_1 \rightarrow \cdots \rightarrow T_N = T$, with each step weighted by a matrix entry of some S_j . By Corollary 5.2 every walk contributes a non-negative summand. The *stay walk* $T_0 = \cdots = T_N = T$ contributes $\prod_k [S_{j_k}]_{T,T}$. Under the descent-0-at-1 and column-descent-free hypotheses, every diagonal factor $[S_{j_k}]_{T,T}$ is one of $\{q+1\} \cup \{[d+1]_q/[d]_q : |d| \geq 2\}$ — all strictly positive in $\mathbb{Q}_{\geq 0}(q)$. The total p_T is bounded below by this stay-walk contribution. $\square$

Combined with Lemma 5.5, the converse direction reduces to:

Proposition 5.7 (Reduction). *If there exists $T^* \in \text{SYT}(\lambda)$ with descent-0 at 1 and no column-descents, then $\text{tr}_{V^\lambda}(\Omega^{(\lambda)}) \neq 0$.*

6 Converse: existence of T^*

We close the converse by giving an explicit construction T_{uniform} of a descent-0-at-1 column-descent-free SYT of shape λ , valid for every $\tau(\widehat{\lambda}) = 0$ shape. The construction is uniform across the regimes $q = 1$ and $q \geq 2$ (in contrast to the earlier case-split version that used the row-superstandard tableau for $q = 1$ and an interleaved construction for $q \geq 2$).

Definition 6.1. An SYT T of shape λ has a *column descent at j* if the entries j and $j+1$ sit in the same column at adjacent rows. T is *column-descent-free* if no j is a column descent.

6.1 The construction T_{uniform}

We use 0-indexed coordinates. Let $L = \ell(\widehat{\lambda})$, $M = |\widehat{\lambda}| - 2L$, $n = 2L + M + q$. Set

$$k := \#\{r \in \{0, \dots, L-1\} : \widehat{\lambda}_r \geq 3\}.$$

Column-0 values. For $r \in \{0, \dots, L+q-1\}$, define

$$v_r := \begin{cases} 2r+1, & 0 \leq r < k, \\ n-2(L+q-1-r), & k \leq r \leq L+q-1. \end{cases} \quad (1)$$

Place $T_{\text{uniform}}(r, 0) := v_r$. These form a band of small odd values $1, 3, \dots, 2k-1$ at the top of column 0 (the “long-row band”) and a band of large odd values $\dots, n-4, n-2, n$ at the bottom (the “tail band”).

Remaining cells. Let $w_1 < w_2 < \cdots < w_{L+M}$ be the remaining values of $\{1, \dots, n\} \setminus \{v_r\}$ in increasing order. Fill the subdiagram $\widehat{\lambda}' := (\widehat{\lambda}_0 - 1, \dots, \widehat{\lambda}_{L-1} - 1)$ row-by-row: row r of $\widehat{\lambda}'$ receives values $w_{a_r+1}, \dots, w_{a_r+\widehat{\lambda}_r-1}$ at columns $1, \dots, \widehat{\lambda}_r - 1$ of λ , where $a_r := \sum_{i < r} (\widehat{\lambda}_i - 1)$.

Example 6.2 ($\widehat{\lambda} = (3, 2)$ at $q = 1$). Here $L = 2$, $M = 1$, $k = 1$, $n = 6$, $\lambda = (3, 2, 1)$. Column-0 values: $v_0 = 1$, $v_1 = 4$, $v_2 = 6$. Remaining $\{2, 3, 5\}$ fill $\widehat{\lambda}' = (2, 1)$ row-superstandard: row 0 receives 2, 3; row 1 receives 5. Resulting tableau:

$$T_{\text{uniform}} = \begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 4 & 5 & \\ \hline 6 & & \\ \hline \end{array}$$

Rows and columns are strictly increasing; entries 1, 2 are in row 0 (descent-0 at 1); no $j, j+1$ pair shares a column.

6.2 Three elementary lemmas

The verification that T_{uniform} has the required properties reduces to three elementary inequalities about v_r and w_i .

Lemma 6.3 (Column-0 gaps). $v_0 = 1$, $v_{L+q-1} = n$, and $v_{r+1} - v_r \geq 2$ for all r .

Lemma 6.4 (Row condition $v_r < w_{a_r+1}$). For every $r \in \{0, \dots, L-1\}$, the column-0 value v_r is strictly less than the first remaining value w_{a_r+1} in row r .

Lemma 6.5 (No column descents). No two consecutive integers in T_{uniform} share a column at adjacent rows.

6.3 The main existence theorem

Theorem 6.6 (Existence of T^*). For every $\widehat{\lambda}$ with $\ell(\widehat{\lambda}) \geq 2$, each row of length ≥ 2 , and every $q \geq 1$ satisfying $\tau(\widehat{\lambda}) = 0$, the tableau T_{uniform} of shape $\lambda = (\widehat{\lambda}, 1^q)$ is a column-descent-free SYT with $T_{\text{uniform}}(0, 0) = 1$ and $T_{\text{uniform}}(0, 1) = 2$.

Proof. By Lemma 6.3, column 0 is strictly increasing. By Lemma 6.4, each row is strictly increasing at the column-0-to-column-1 transition; strict increase within $\widehat{\lambda}'$ follows from the increasing w -sequence and row-major filling. Column strictness within $\widehat{\lambda}'$ likewise. Hence T_{uniform} is a valid SYT. The values $T_{\text{uniform}}(0, 0) = v_0 = 1$ and $T_{\text{uniform}}(0, 1) = w_1 = 2$ are immediate. By Lemma 6.5, no column descents. $\square$

Corollary 6.7 (Converse direction). If $\tau(\widehat{\lambda}) = 0$, then $\text{tr}_{V^\lambda}(\Omega^{(\lambda)}) \neq 0$.

Proof. Combine Theorem 6.6 with Proposition 5.7. $\square$

Together with Corollary 4.4, this closes the proof of Theorem 1.1.

7 Sharpness at $\tau + 1$

Pillar 1 (Theorem 4.1) places ΩV^λ inside $\bigcap_{j=1}^\tau E_j^-$. A natural question is whether this depth is best possible. We prove that it is, in the strong form $B^{(\lambda)} \cap E_{\tau+1}^- = \{0\}$.

7.1 Statement

Write $B^{(\lambda)} := \Omega^{(\lambda)} V^\lambda$ for the chain image.

Theorem 7.1 (Strong sharpness). Under the standing hypothesis ($\ell(\widehat{\lambda}) \geq 2$, every row ≥ 2 , $q \geq |\widehat{\lambda}| - 2\ell(\widehat{\lambda}) + 2$),

$$B^{(\lambda)} \cap E_{\tau+1}^-|_{V^\lambda} = \{0\},$$

equivalently, $S_{\tau+1} := T_{\tau+1} + 1$ is injective on $B^{(\lambda)}$.

Corollary 7.2. $\sup\{j : B^{(\lambda)} \subseteq E_j^-|_{V^\lambda}\} = \tau(\widehat{\lambda})$ exactly.

Theorem 7.1 is proved in [19]; combined with Pillar 1 [15], it gives the exact descent depth in Corollary 7.2.

7.2 The one-fact proof

The proof rides on a single structural fact about the outer product $\mathcal{O} := \prod_{j=\ell}^{n-2} (T_j + 1)$ appearing in the Pillar 1 reduction theorem, together with a geometric commutativity observation.

Lemma 7.3 (Outer product injectivity). *$\mathcal{O}$ is injective on $E_{n-1}^+|_{V^\lambda}$, with image in $E_\ell^+|_{V^\lambda}$.*

The proof is a chain application of the adjacent-injectivity identity $E_j^- \cap E_{j+1}^+ = 0$: for $v \in E_{n-1}^+$, $(T_{n-2} + 1)v \in E_{n-2}^+$ and the restriction $(T_{n-2} + 1)|_{E_{n-1}^+}$ is injective; iterating down to $j = \ell$ gives the claim.

Lemma 7.4 (Commutativity). *Under the standing hypothesis, $S_{\tau+1}$ commutes with every factor of $\mathcal{O}$ and with each contributor Phi-embedding $\Phi_{k,*}$.*

Proof sketch. The key inequality is

$$\ell - (\tau + 1) = |\widehat{\lambda}| - \ell(\widehat{\lambda}) \geq \ell(\widehat{\lambda}) \geq 2,$$

using every-row- ≥ 2 and $\ell(\widehat{\lambda}) \geq 2$. Hence every factor index $j \geq \ell$ of $\mathcal{O}$ satisfies $|j - (\tau + 1)| \geq 2$, and Hecke generators at distance ≥ 2 commute. The Phi-embeddings are T_i -equivariant for $1 \leq i \leq n - 3$, and the same inequality places $\tau + 1$ in this range. $\square$

Proof of Theorem 7.1 (sketch). Strong induction on $|\widehat{\lambda}|$. Let $w \in B^{(\lambda)}$ with $S_{\tau+1}w = 0$. By the Pillar 1 reduction theorem, $w = \mathcal{O} \sum_k \Phi_{k,*}(u_k)$ uniquely with $u_k \in B^{(\mu_{k,*})}$. By Lemma 7.4, $S_{\tau+1}$ commutes with $\mathcal{O}$ and with each $\Phi_{k,*}$, so

$$0 = S_{\tau+1}w = \mathcal{O} \sum_k \Phi_{k,*}(S_{\tau+1}u_k).$$

The inner sum lies in $E_{n-1}^+|_{V^\lambda}$; by Lemma 7.3 each $\Phi_{k,*}(S_{\tau+1}u_k) = 0$, and disjoint Phi-supports plus injectivity of each $\Phi_{k,*}$ give $S_{\tau+1}u_k = 0$. By the threshold-matching identity, $\tau(\hat{\mu}_{k,*}) = \tau(\widehat{\lambda})$ for every contributor, so the IH at $\mu_{k,*}$ gives $u_k = 0$, hence $w = 0$. $\square$

7.3 Hook base case

The induction bottoms out on the hook layer $\widehat{\lambda} = (k_h)$ (length 1 shapes, formally outside the standing hypothesis but reached as contributor inner shapes). The hook analogue is

Lemma 7.5 (Hook sharpness). *For $\mu = (k_h, 1^{m_h})$ with $k_h \geq 2$ and $m_h \geq k_h$, $S_{m_h-k_h+2}$ is injective on $B^{(\mu)}$.*

Proof sketch. Induction on k_h with base case $k_h = 2$ by direct Hoefsmit computation: $B^{(2,1^{m_h})} = E_{m_h+1}^+|_{V_\mu}$ is a line, spanned by $w_0 = v_{T^+} + \beta v_{T^-}$ with T^+ placing the top two values at corners $(1, 2)$ and $(m_h + 1, 1)$ and T^- swapping them. In T^+ , the cells of m_h and $m_h + 1$ are non-adjacent (axial distance $m_h \neq \pm 1$), giving a non-degenerate T_{m_h} -block with nonzero off-diagonal; in T^- , same column, so $S_{m_h}v_{T^-} = 0$. The image $S_{m_h}w_0$ therefore has a nonzero coefficient on a basis vector outside $\text{supp}(B^{(\mu)})$, witnessing injectivity. The inductive step uses a hook analogue of the Pillar 1 reduction with $\mathcal{O}^{(\mu)} = \prod_{j=m_h+1}^{m_h+k_h-2} (T_j + 1)$ and contributor $\mu_{1,*} = (k_h - 1, 1^{m_h-1})$, the same commutativity argument as above, and the IH on k_h . $\square$

7.4 Verification

The full rank $\dim S_{\tau+1}(B^{(\lambda)}) = \dim B^{(\lambda)}$ has been verified on 14 datapoints (eight multi-row shapes and six hooks) computed via a sparse Hoefsmit implementation mod $P = 100003$ at $q = 11$; see [19, Table 1].

8 Per-SYT diagnostics

The converse direction (Section 5) was proved by showing each diagonal $p_T(q) := [v_T](\Omega v_T)$ lies in the positive cone $\mathbb{Q}_{\geq 0}(q)$ and exhibiting a non-vanishing T_{uniform} . In this section we collect three independent invariants that pin down which T 's contribute and which do not, framing the trace dichotomy as a confluence of structural diagnostics rather than a single identity.

8.1 Diagnostic 1: per-SYT positivity

Theorem 8.1 (Per-SYT positivity, restated). *For every $T \in \text{SYT}(\lambda)$, the diagonal $p_T(q) \in \mathbb{Q}_{\geq 0}(q)$ in lowest terms.*

This is Theorem 5.3, immediate from Hoefsmit positivity (Lemma 5.1) and closure of the cone under sums and products. Combined with the existence of T_{uniform} (Theorem 6.6), it closes the converse: $\tau(\hat{\lambda}) = 0$ forces $p_{T_{\text{uniform}}}(q) \neq 0$ on a stay-walk lower bound, so the trace is non-vanishing. The cleavage is structural — there is no fortuitous cancellation between T -summands, since none can cancel against another in a positive cone.

8.2 Diagnostic 2: leftmost-factor SC vanishing

Theorem 8.2 (Per-SYT leftmost-factor sign-kill). *Let $\lambda \vdash n$ with $\ell = \ell(\lambda)$. If the values ℓ and $\ell+1$ sit in the same column of T (pair $(\ell, \ell+1)$ is SC at T , equivalently $S_{\ell}v_T = 0$), then $p_T(q) = 0$.*

Two-line proof. Write $\Omega = R'_{\ell+1}R'_{\ell+2} \cdots R'_n = S_{\ell} \cdot R'_{\ell}R'_{\ell+2} \cdots R'_n =: S_{\ell} \cdot X$. Then $\Omega v_T = S_{\ell}(Xv_T) \in \text{im}(S_{\ell}) = E_{\ell}^+|_{V^{\lambda}}$, while $v_T \in E_{\ell}^-|_{V^{\lambda}}$ by SC. Project the seminormal expansion $\Omega v_T = \sum_{T'} c_{T,T'} v_{T'}$ onto E_{ℓ}^- along E_{ℓ}^+ : the left side is 0, the right side has $c_{T,T''} = 0$ for every $T'' \in \text{SYT}^{\text{SC at } \ell}$; specialise $T'' = T$. $\square$

Remark 8.3. The theorem holds without any restriction on λ (multi-row, τ , staircase). It is the per-SYT refinement of the universal containment $B^{(\lambda)} \subseteq E_{\ell}^+|_{V^{\lambda}}$ obtained from the same leftmost-factor identity. See [20] for the full exposition.

At $\lambda = (3, 3, 1, 1, 1)$, an empirical sweep produced 108 zero diagonals out of $|\text{SYT}(\lambda)| = 120$. Theorem 8.2 accounts for 26 of these — exactly the descent-0 SYTs with pair $(5, 6)$ same-column. In particular, the historically mysterious zero at

$$T_0 = ((1, 2, 5), (3, 4, 6), (7), (8), (9))$$

is explained by SC at $(5, 6)$ (value 5 at row 0 column 2, value 6 at row 1 column 2).

8.3 Diagnostic 3: Pillar-1-side SR vanishing

The Pillar-1 containment $B^{(\lambda)} \subseteq E_j^-$ for $j = 1, \dots, \tau$ has a per-SYT shadow obtained by the same projection argument run at index $j \leq \tau$:

Corollary 8.4 (Pillar-1-side per-SYT vanishing). *If $T \in \text{SYT}(\lambda)$ has pair $(j, j+1)$ same-row for some $j \in \{1, \dots, \tau\}$ (i.e., $v_T \in E_j^+$), then $p_T(q) = 0$.*

Theorems 8.2 and Corollary 8.4 provide *three* independent invariants pinning τ : structural depth (Pillar 1 + sharpness), trace vanishing (Theorem 1.1), and per-SYT leftmost/rightmost diagnostics. Any future categorical lift (Conjecture 10.1) must unify all three.

8.4 Open: companion diagnostic and interior zeros

SR companion conjecture. Theorem 8.2 fires at the ℓ -side via SC. A mirror statement is expected at the τ -side: *if pair $(j, j+1)$ is same-row at T for some $j \in \{1, \dots, \tau\}$, then $p_T(q) = 0$.* This is exactly Corollary 8.4 above, so the “mirror” is in fact already proved. A finer companion conjecture concerns the rightmost-factor identity $\Omega = Y \cdot S_1$ extended to inner positions: a one-page proof is queued in the agent’s proof-attempts file.

Interior D-class zeros. The $108 - 26 = 82$ zero diagonals at $\lambda = (3, 3, 1, 1, 1)$ not explained by Theorem 8.2 split further: 14 are descent-1 trivial vanishings (pair $(1, 2)$ SC, $\Omega v_T = 0$ outright), and 68 are descent-0 SYTs with pair $(\ell, \ell+1)$ diagonal (D-class). For these, v_T has nonzero components in both E_ℓ^+ and E_ℓ^- , and the seminormal projection no longer forces $c_{T,T} = 0$ directly. These “interior cancellations” are not explained by any of the three diagnostics above; characterising them is the natural next-horizon refinement of per-SYT analysis.

9 Computational verification

The four ingredients have been verified independently.

Scope. The dichotomy was tested on every shape $\lambda = (\hat{\lambda}, 1^q)$ with $|\lambda| \leq 25$ satisfying the hypothesis $\ell(\hat{\lambda}) \geq 2$ and every row ≥ 2 , yielding 5957 shapes in total: 210 with $|\lambda| \leq 14$; 832 with $|\lambda| \leq 18$; 2688 with $|\lambda| \leq 22$; 5957 with $|\lambda| \leq 25$.

Methodology.

- Symbolic computation in `sympy` verified Hoefsmit positivity (Lemma 5.1) for axial distances $|d| \leq 6$.
- Direct construction confirmed the T_{uniform} algorithm produces column-descent-free SYTs on all 5957 shapes (`t_unified.py`).
- Pillar 1 and trace-vanishing were verified via sparse application of $\Omega^{(\lambda)}$ to random vectors over $\mathbb{F}_p$ with $p = 100003$, $q = 11$, with rank stability checks across multiple probes.

Refuted alternatives. Three structurally different routes were investigated and ruled out by computational data:

- A q -binomial product ansatz for $\text{tr}(\Omega)$ failed at $\lambda = (4, 3, 1^2)$.
- The non-crossing matching factorisation (motivated by Jucys–Murphy products) gave rank 12 where Catalan-counting predicted 5.
- The Goertzen–Williamson variational characterisation [13] of the seminormal basis is only sharp at $v = 1$ (Soergel calculus specialisation), giving no generic- q positivity.

- The Bai–Gu–Guo–Liu 1423-avoidance criterion [14] matches our diagonals on 7/7 tested shapes, but the underlying rings differ (theirs is $\mathbb{Z}_{\geq 0}[\beta]$ pipe-dream counts; ours is $\mathbb{Q}_{\geq 0}(q)$ with cyclotomic poles).

Remark 9.1. Ram’s content multiset formula [5] and the Tolmachov–Zhylynskyi closed form for $\text{tr}(\prod p_i(J_i))$ [6] suggest a route via Jucys–Murphy products. However, the multiset $\{c_i(T) : i\}$ is a *shape* invariant, so symmetric JM expressions cannot detect a $\hat{\lambda}$ -vs- q dichotomy. Trace-vanishing of $\Omega^{(\lambda)}$ requires a position-asymmetric / non-central JM expression; the right-absorption proof bypasses this entirely.

10 Outlook: a categorical lift

The forward direction of Theorem 1.1 proved strictly more than trace vanishing: it gave $\Omega^{(\lambda)2} = 0$ on V^λ , with the matrix of $\Omega^{(\lambda)}$ in the descent-at-1 block decomposition having the form $\begin{pmatrix} 0 & 0 \\ X & 0 \end{pmatrix}$. This structural rigidity invites a categorification.

10.1 The conjecture

Conjecture 10.1 (Categorical dichotomy). *Let $\tilde{\Omega}^{(\lambda)}$ be a Rouquier complex categorifying $\Omega^{(\lambda)}$ as an object of the homotopy category of Soergel bimodules $K^b(\text{SBim}_n)$, and let P_λ denote Hogancamp’s categorified Young symmetrizer for λ . Then*

$$\text{tr}_{V^\lambda}(\Omega^{(\lambda)}) \equiv 0 \iff \text{HH}_*(P_\lambda \cdot \tilde{\Omega}^{(\lambda)} \cdot P_\lambda) = 0$$

in bidegree ($q\text{-deg} = |\lambda|$, $\text{hom-deg} = 0$).

The Specht-isotypic projector P_λ realises V^λ as the λ -component of the universal Markov trace; the conjecture says that the $\hat{\lambda}/\tau$ dichotomy survives intact at the categorical level.

10.2 What the lift should achieve

Specht-isotypic decomposition. The honest gap is that the Specht-isotypic decomposition

$$\text{Tr}(\text{SBim}_n) = \bigoplus_{\lambda} \text{Tr}^\lambda$$

of the horizontal trace into derived blocks is not yet explicitly written down in the derived setting [7]. Hogancamp’s categorified Young symmetrizers [8, 9] provide P_λ , and the conjectural decomposition should be $\text{Tr}^\lambda = \text{Tr}(\text{SBim}_n \cdot P_\lambda)$.

Lifting right-absorption. The identity $S_1^2 = (q+1)S_1$ should categorify to an explicit homotopy / contractibility statement in the categorified bar complex. This is the categorical analogue of “ Ω ends in S_1 , and $S_1^2 = (q+1)S_1$ ” — the load-bearing identity behind Lemma 4.2.

10.3 Reading list

Reading list, in suggested order, for tackling Conjecture 10.1:

- (1) [11] ([arXiv:2507.10061](https://arxiv.org/abs/2507.10061)) — Idempotents, traces, and dimensions in Hecke categories. Sits at the idempotent / trace junction; may already contain a Specht-isotypic decomposition.

- (2) Trinh–Williams [10] ([arXiv:2601.17293](#)) — Partial Springer resolutions and Hecke trace formulas.
- (3) Hogancamp [8] ([arXiv:1505.08148](#)) and Abel–Hogancamp [9] ([arXiv:1510.05330](#)) — Categorified Young symmetrizers.
- (4) Gorsky–Hogancamp–Wedrich [7] ([arXiv:2002.06110](#)) — Derived horizontal trace of Soergel bimodules.
- (5) Wedrich’s 2025 survey [12] ([arXiv:2509.08478](#)) for context on the link-homology / TQFT environment.

10.4 The Demazure–Lusztig bridge

A second next-horizon direction is the decategorified bridge between the LR / Schubert structure constants and the Markov-trace evaluation, mediated by the Demazure–Lusztig operator on $K_T(G/B)$. Five published papers [21, 22, 23, 24, 25] use this operator to evaluate equivariant K -theory pairings on the flag variety in terms of Hecke-algebra characters. The trace $\mathrm{tr}_{V^\lambda}(\Omega^{(\lambda)})$ should equal a localisation residue on the λ -fixed point of $K_T(G/B)$ via Atiyah–Bott / AMSS, and $\tau(\hat{\lambda}) \geq 1$ should manifest as residue cancellation in this localisation expansion. This is the natural next-horizon for a *structural* (as opposed to algebraic) understanding of τ : the threshold appears here as the height of a torus-fixed point inside a Schubert stratification, not as a combinatorial count. Working out exactly which Demazure–Lusztig word evaluates to $\Omega^{(\lambda)}$ and extracting the residue cancellation conjecturally encoded by $\tau \geq 1$ is left for future work.

10.5 What this paper does not address

We do not address the non-stable regime $q < |\hat{\lambda}| - 2\ell(\hat{\lambda}) + 2$ where the standing hypothesis fails, or refinement of Theorem 5.3 to $p_T(q) \in \mathbb{Z}_{\geq 0}[q]$ (the trace is empirically a non-negative-coefficient polynomial, but our Hoefsmit-based proof gives only rational positivity); see Remark 3 of [17]. The interior D-class zero diagonals at $\lambda = (3, 3, 1, 1, 1)$ (Section 8.4) remain unexplained by the three diagnostics collected here.

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